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Term Project: Instant Messaging App

Feature Document

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1 Introduction

This document describes the proposed features and requirements for a frontend-focused instant messaging application. The application is closely related in concept to popular messaging platforms such as Slack, Discord, and WhatsApp, but is limited to essential messaging and chat room functionality. This document serves as a foundation for agile planning, system design, and future development activities.

1.1 Purpose

The purpose of this document is to clearly define the product vision, scope, user needs, and feature requirements for the instant messaging application. This feature document helps guide my development decisions, supports sprint planning, and ensures that the final product aligns with user expectations and project constraints. It will be detailed enough to the point where someone can take this document and work on the app features themselves with limited contact to me.

1.2 Target Audience

The target audience for this document is:

- Product Manager (TA)
- Other developers working on the application
- Myself

1.3 Terms and Definitions

- **UI**: User Interface
- **Frontend**: The client-side portion of the application, responsible for user interaction
- **Backend**: Server-side services accessed through APIs (treated as a black box in this project)
- **Chat Room**: A virtual space where users exchange messages
- **Epic**: A large user story that can be broken down into smaller user stories

- **User Story**: A short description of a feature from the user's perspective

2 Product Overview

This section provides an overview of the product and identifies who it is designed for. The product is an easy-to-use instant messaging application that enables real-time communication between users. It is intended for individuals or small groups who need fast, reliable messaging for collaboration or social interaction. The goal of this product is to simplify group communication by providing essential messaging features without the complexity of larger messaging platforms.

2.1 Product Vision

For users who want a simple and intuitive way to communicate in real time **Who** need an easy-to-use messaging interface without unnecessary complexity. **The** Instant Messaging Application **Is a** web-based chat application **That** allows users to authenticate, create chat rooms, and exchange messages in real time **Unlike** full-scale messaging platforms such as Slack or Discord **Our product** focuses on essential messaging features with a clean, beginner-friendly, simple user experience.

2.2 Product Scope

This section describes the scope of the instant messaging application by outlining all major features provided by the frontend and backend. The scope is visualized using a mind map that illustrates the core functional areas of the system and their relationships as seen below:

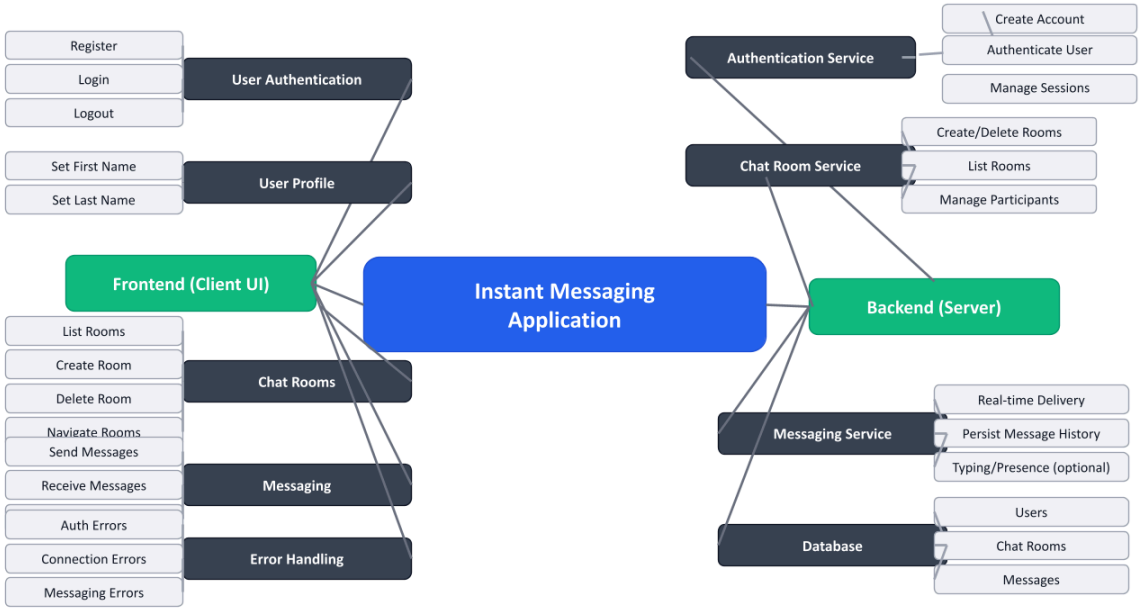

Figure 1. Product Scope Mind Map (Frontend + Backend)

2.3 Operating Environment

The application will run in a modern web browser on desktop or laptop computers. Users will access the application through a web-based interface. The frontend will communicate with a provided backend service over the Internet using HTTP requests and real-time communication protocols.

2.4 Design and Implementation Constraints

The frontend will be implemented using modern web technologies such as:

- **React** for building the user interface
- **JavaScript/TypeScript** for application logic
- **HTML and CSS** for layout and styling

The backend is provided as a black-box service and is accessed only through documented APIs. No direct access to backend source code or database is available. This constraint

limits implementation to client-side features and requires careful handling of API responses and error conditions.

3 User Stories

The purpose of this section is to describe the users of the instant messaging application and to understand how they might interact with the system. Understanding users is essential in software product development because features should be designed to support real user needs rather than assumptions made by developers. The more detailed a story is, the more clear a feature is.

3.1 Personas

Persona 1: Alex – College Student

- Alex is a 20-year-old computer science student who frequently works on group projects with classmates. He is comfortable using modern web applications and messaging platforms. Alex values fast communication and the ability to organize conversations by topic. He wants a simple messaging application that allows him to participate in multiple chat rooms without unnecessary complexity.

Persona 2: Jordan – Teaching Assistant

- Jordan is a 26-year-old graduate student working as a teaching assistant. Jordan uses messaging tools to coordinate with students and manage discussions for different courses. While technically competent, Jordan prefers tools that are easy to navigate and reduce the effort required to manage group communication. Jordan is interested in features such as chat room creation and clear conversation organization. Jordan also prefers simple UI so their students don't have troubles adjusting.

Persona 3: Sam – Casual User

- Sam is a 40-year-old casual user who uses messaging applications occasionally for small group conversations. Sam has basic technical skills and prefers intuitive interfaces. Sam is primarily interested in simple messaging functionality without advanced or distracting features. Mainly uses with family and friends.

3.2 User Scenarios

3.2.1 User Scenarios with Persona 1

Scenario 1: Coordinating a Software Engineering Group Project

Alex is a computer science student enrolled in a software engineering course that requires a term-long group project. His group members have different schedules, and they rarely have time to meet in person. Alex logs into the instant messaging application from his laptop after classes to check for updates from his teammates. After authentication, he views a list of chat rooms and selects the one dedicated to his project group.

Alex scrolls through the conversation history to catch up on messages sent earlier in the day, including updates about task assignments and design decisions. As he reviews the discussion, he gains a clearer understanding of what work still needs to be completed. Alex then sends a message asking a clarifying question about a task he was assigned and waits for responses. As messages arrive in real time, Alex replies and adjusts his plan based on the group's feedback. The chat room allows Alex and his teammates to stay aligned on project goals without relying on email or external tools.

Scenario 2: Managing Multiple Academic Conversations

During the middle of the term, Alex is involved in several group-based courses at the same time. Each course has its own group discussions, and Alex needs a way to keep conversations organized. He logs into the application and views the list of available chat rooms, each labeled according to a different class or purpose.

Alex switches between chat rooms as he works through his assignments for the evening. In one room, he participates in a discussion about a programming assignment. In another room, he reviews messages related to exam preparation for a different course. By separating conversations into distinct chat rooms, Alex avoids confusion and can easily

focus on the work that needs to be done now. The ability to navigate between rooms helps Alex manage his workload more efficiently and reduces the mental overhead of tracking multiple conversations.

Scenario 3: Creating and Cleaning Up Study Spaces

As an exam approaches, Alex decides to create a temporary study group with classmates. He logs into the application and creates a new chat room specifically for exam review. After the room is created, classmates join and begin sharing study tips, practice problems, and reminders about exam topics. Alex regularly checks the chat room to stay engaged with the group and contribute to discussions.

Once the exam has passed, the study group is no longer active. Alex logs back into the application and notices that the chat room is no longer being used. To keep his interface clean, he deletes the study group chat room. This helps Alex maintain a clean list of active conversations and focus only on chat rooms that are currently relevant.

3.2.2 User Scenarios with Persona 2

Scenario 1: Supporting Students Outside of Class

Jordan is a teaching assistant for a large undergraduate course and frequently receives repetitive questions from students via email. To streamline communication, Jordan logs into the instant messaging application and creates a chat room dedicated to the course discussion section. The goal is to provide a shared space where students can ask questions and see responses that may also help others.

Throughout the week, Jordan checks the chat room during breaks and in the evenings. Before responding to new questions, Jordan reads earlier messages to understand what has already been discussed. By answering questions in the chat room, Jordan reduces duplicated effort and ensures that all students have access to the same information. The chat room becomes a central hub for course-related communication.

Scenario 2: Managing the Lifecycle of Course Discussions

As the course progresses, different assignments and exams require focused discussion for limited periods of time. After an assignment deadline passes, Jordan notices that the corresponding chat room is no longer active. To avoid confusion and prevent students from posting outdated questions, Jordan deletes the inactive chat room.

By removing unused discussion spaces, Jordan ensures that students can easily identify current and relevant chat rooms. This organization helps maintain clarity and improves the overall communication experience for both students and instructors.

3.2.3 User Scenarios with Persona 3

Scenario 1: Participating in a Short-Term Group Conversation

Sam occasionally uses messaging applications to communicate with small groups of friends. Sam creates an account on the instant messaging application and logs in to join a chat room created for planning a social event. Upon entering the chat room, Sam reads through previous messages to understand what has already been decided.

Sam sends a few messages to share availability and opinions, then continues to receive responses from other participants. The conversation happens naturally and does not require Sam to learn complex features or settings. After the planning is complete, Sam logs out of the application, satisfied that the communication was quick and straightforward.

3.3 Epic User Stories

3.3.1 Epic User Stories 1

Epic 1: User Authentication and Profile Setup

As a user ***I want to*** register, log in, log out, and set my profile information ***So that I can*** securely access the messaging application and be identified by other users.
Acceptance Criteria ***Given*** a user wants to access the application ***When*** the user registers or logs in with valid credentials, ***Then*** the user is authenticated and granted access to messaging features.
User story 1: As a new user, I want to register for an account so that I can access the application.
User story 2: As an authenticated user, I want to log out so that my account remains secure.
User story 3: As an authenticated user, I want to set or update my profile name so that others can identify me.

3.3.2 Epic User Stories 2

Epic 1: Chat Room Management

As a student ***I want to*** I want to create, view, navigate, and remove chat rooms, ***So that*** conversations can be organized by group or topic.
Acceptance Criteria ***Give*** an authenticated student or moderator, ***When*** the user creates or deletes a chat room, ***Then*** the chat room list is updated accordingly.
User story 1: As a student, I want to see a list of available chat rooms so that I can choose where to participate.
User story 2: As a student, I want to create a chat room so that I can start a new conversation.
User story 3: As a student, I want to delete chat rooms that are no longer needed so that my interface stays organized.

3.3.3 Epic User Stories 3

Epic 1: Real-Time Messaging and Participation
As a chat participant ***I want to*** send and receive messages and view conversation history, ***So that I can*** effectively communicate with others in real time.

Acceptance Criteria
Given a casual user is participating in a chat room, ***When*** messages are sent or received, ***Then*** messages appear in the conversation in real time.
User story 1: As a casual user, I want to send messages so that I can communicate with others in the chat room.
User story 2: As a casual user, I want to view message history so that I can understand past discussions.
User story 3: As a casual user, I want to see error messages when something goes wrong so that I know how to respond.

4 Feature List

This section provides a comprehensive list of features included in the instant messaging application. The features are organized by priority. Each feature represents an independent and coherent unit of functionality that can be implemented within a sprint. Effort estimates are based on frontend development only.

Feature 1: User Registration	**Priority: High**	**Effort Required:** 1-2
Description: Allows a new user to create an account by providing required credentials in order to access the application.		
Constraints: Depends on the availability of backend authentication APIs.		
Comments: Core feature required before accessing any other functionality.		

Feature 2: User Login	**Priority: High**	**Effort Required: 1.5**
Description: Allows a registered user to authenticate and access the messaging interface.		
Constraints: Requires successful communication with backend authentication services.		
Comments: Must handle invalid credentials and authentication errors.		

Feature 3: User Logout	**Priority: Medium**	**Effort Required: 1**
Description: Allows an authenticated user to securely end their session and return to the login screen.		
Constraints:		

Session handling is managed through backend APIs.
Comments: Improves security and session control.

Feature 4: Profile Name Management	**Priority: Medium**	**Effort Required: 1**
Description: Allow Users to set or update their first and last name for display within chat rooms.		
Constraints: Profile updates depend on backend data storage.		
Comments: Used to identify users in conversations.		

Feature 5: View Chat Room List	**Priority: High**	**Effort Required: 1**
Description: Displays a list of available chat rooms that the user can join or navigate between.		
Constraints: Chat room data is retrieved from backend services.		
Comments: Primary navigation feature for the application.		

Feature 6: Create Chat Room	**Priority: High**	**Effort Required: 2**
Description: Allows a user to create a new chat room for group or topic-based conversations.		
Constraints: Requires backend support for chat room creation.		
Comments: Frequently used by student users and moderators.		

Feature 7: Navigate Between Chat Rooms	**Priority: High**	**Effort Required: 1**
Description: Allows users to switch between different chat rooms without leaving the application.		
Constraints: Must preserve message context when switching rooms.		
Comments: Essential for users participating in multiple conversations.		

Feature 8: Delete Chat Room	**Priority: Medium**	**Effort Required: 1**
Description: Allows users to remove chat rooms that are no longer needed.		
Constraints: Requires backend authorization and confirmation handling.		
Comments: Used to reduce clutter and manage inactive rooms.		

Feature 9: Send Messages	**Priority: High**	**Effort Required: 2**
Description: Allows users to send messages within a selected chat room.		
Constraints: Depends on backend real-time messaging support.		
Comments: Core functionality of the application.		

Feature 10: Receive Messages (Real Time)	**Priority: High**	**Effort Required: 2**
Description: Displays incoming messages from other users in real time within the active chat room.		

Constraints: Requires reliable network connection and backend message delivery.
Comments: Critical for usability and responsiveness.

Feature 11: Display Message History	**Priority: Medium**	**Effort Required: 1.5**
Description: Displays previously sent messages when a user enters a chat room.		
Constraints: Message history retrieval depends on backend data availability.		
Comments: Provides context for ongoing conversations.		

Feature 12: Error Message Display	**Priority: Low**	**Effort Required: 1**
Description: Displays user-friendly error messages for authentication failures, connection issues, or messaging errors.		
Constraints: Requires consistent error responses from backend services.		
Comments: Improves user experience and clarity.		